

CHOLLETI DEEKSHITHA CHARVY

Computer Science Student | Data Structures & Algorithms | Problem Solving | Web Development

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SKILLS

Programming Languages :	C, Java, Python
Frameworks & Libraries :	HTML, CSS, JavaScript, Node.js, Express.js
Tools & Platforms :	VS Studio, Jupyter Notebook, Google Collab
Databases :	MongoDB
Soft Skills :	Communication, Teamwork, Problem Solving, Time Management, Critical Thinking
Languages :	English, Telugu, Hindi

PROJECTS

Football Analysis Project | Github

- Developed an **AI-based football analytics system** to detect and track **players**, referees, and the ball in match videos using the **YOLOv5** deep learning model.
- Implemented **computer vision pipelines** with **OpenCV**, **NumPy**, and **Pandas** to process video frames, analyze player movements, and extract structured match data.
- Applied **K-Means clustering** to segment jersey colors and classify players into teams based on t-shirt color detection.
- Integrated **optical flow algorithms** to estimate camera motion and correct positional tracking for accurate player movement analysis.
- Used **perspective transformation** to convert pixel-based positions into real-world measurements, enabling calculation of player speed, distance covered, and team ball possession statistics.
- Trained and optimized the **YOLO model** for football match scenarios, improving detection performance for players, referees, and the ball.

Technologies / Tools Used : Python, NumPy, Pandas, Matplotlib, openCV, ultralytics, supervision

Simple_BrainTumor_Detection | Github

- Developed a **web-based medical imaging application** to detect **brain tumors** from CT/MRI scans using the **YOLOv8** deep learning model and an interactive interface with **Streamlit**.
- Implemented tumor localization with a pre-trained YOLO model to identify abnormal regions and generate **bounding boxes** with confidence scores for accurate detection.
- Integrated a **K-Nearest Neighbors (KNN)** classifier from **scikit-learn** to classify tumors based on size, analyzing bounding box area to categorize them into small and large tumor types.
- Built an image processing pipeline using **OpenCV**, **Pillow**, and **NumPy** for image preprocessing, color space conversions, and numerical computations.
- Designed a user-friendly interface for users to upload medical images (JPG/PNG), visualize tumor detection results with bounding boxes, and view prediction metrics including tumor coordinates, classification, and confidence levels.

Technologies / Tools Used : Ultralytics, Scikit-learn (KNN), OpenCV (cv2), Pillow (PIL), NumPy

EDUCATIONS

Anurag University

Bachelor of Technology (B.Tech)

Aug 2024 - Present

CGPA : 9.35/10

- Bachelor of Technology (B.Tech) in Computer Science and Engineering;** Expected Graduation: 2028

Narayana Junior College

Intermediate Board Of Education

Percentage : 98.1

CERTIFICATIONS

Programming In Java

NPTEL

Certification Url : https://drive.google.com/file/d/1FgQH2PxAXU3frgetZr_-WKMchN_a82HO/view?usp=sharing

Generative AI for Developers

cloudskillsboost by Google

Certification Url : https://www.skills.google/public_profiles/5b35b5cb-527c-457c-9d06-481a8046f659

OBJECTIVE

Computer Science undergraduate seeking a **Software Engineering Internship** to apply **programming skills** in **Java** and **Python** for developing scalable and efficient software solutions. Interested in collaborating with engineering teams, learning modern development practices such as **Agile** and **CI/CD**, and contributing to innovative technologies addressing real-world challenges.